FAISAL SHAKEEL

Data Analyst

03096531868 faisalshakeel445@gmail.com www.linkedin.com/in/faisal445

PROFESSIONAL SUMMARY

Data Analyst with experience of building automated RAM (Road Asset Management) pipelines, analytics workflows, and GIS based reporting systems for large-scale transportation and infrastructure projects. Specialized in transforming raw road survey videos and images into structured datasets, condition indexes (ACI/PCI), and actionable geospatial insights. Strong background in Computer Vision and AI, with hands-on expertise in training YOLO-based detection, segmentation, and tracking models to automate road asset identification at scale. Experienced in data engineering, reporting automation, and leading annotation teams to deliver high quality datasets for national-level RAM projects.

PROFESSIONAL EXPERIENCE

Airloop - Data Analyst (Pavement Condition Index & Asset Condition Index)

Oct 2024 – Present | Transportation Asset Management & ACI Reporting

Core Role: Data Analysis

- Performed comprehensive data analysis on road asset condition datasets to accurately compute Asset Condition Index (ACI), including Pavement Condition Index (PCI), to support maintenance decision-making
- Designed and implemented a semi-automated workflow for ACI calculation using survey data, significantly improving efficiency and reducing manual effort
- Developed basic AI models for automated detection of road assets (traffic signs, lane markings, pavement markings), reducing overall project turnaround time by up to 40%
- Built automated Python workflows to convert raw roadway survey videos and images into structured CSV datasets and ACI-driven reports
- Cleaned and organized large volumes of geospatial asset data to ensure accurate and reliable infrastructure information
- Integrated datasets with ArcGIS Pro to create geodatabases (GDBs), maps, and spatial reports for improved visualization and analysis
- Automated reporting processes to reduce manual effort and enable faster asset maintenance and planning decisions

Computer Vision & AI Contributions

- Trained and optimized YOLOv11 detection, segmentation, and tracking models to automatically identify roadway assets from video data
- Automated extraction of traffic signs, poles, manholes, lane markings, guardrails, and other assets, significantly improving processing speed and scalability
- Applied data augmentation, class balancing, and preprocessing techniques to improve model robustness across varying lighting, weather, and road conditions
- Led and mentored the annotation team, implemented quality control checks, and ensured high-quality labeled datasets for reliable model training

Key Project – KSA Project 001 (National-Level Initiative)

- Delivered an end-to-end data analytics and AI solution for nationwide roadway asset assessment, generating structured datasets, ACI condition indices, and GIS dashboards to enable efficient asset monitoring and maintenance planning.
- Developed detection, segmentation, and tracking pipelines connected with automated reporting
- Managed multiple asset categories independently, from data preparation to final client delivery

Data Extraction Operator (Pavement Condition Index & Asset Condition Index)

Transconomy | Oct 2023 – Sep 2024

- Contributed to projects implementing roadway asset detection and assessment solutions for traffic signs, pavement markings, lane markings, traffic signals, and linear assets.
- Worked on Pavement Condition Index (PCI) evaluation, detecting cracks and surface distress for pavement performance assessment (aligned with ASTM D6433).
- Performed detection and segmentation annotations to support AI training datasets.
- Awarded Best Employee of the Quarter for high-quality annotations and contributions to AI-based transportation projects.

PROJECTS

- **YOLO Detection Models:** Custom YOLOv11 models trained for detecting traffic signs, pavement markings, light poles, manholes, catch basins, and gantries.
- **Segmentation Models:** Developed segmentation models for lane markings, curbstones, footpaths, guardrails, and New Jersey barriers.
- **Object Tracking Pipelines:** Designed workflows to track roadway assets across video sequences for consistency and condition monitoring.
- **Pavement Surface Analysis (PCI):** Assisted in detecting cracks and surface distress for pavement condition indexing (ASTM D6433).
- **GIS Integration:** Processed CSV datasets from AI pipelines and integrated with ArcGIS Pro to generate GDBs, interactive maps, and asset visualizations.

TECHNICAL SKILLS

- **Computer Vision & AI Models:** YOLOv11 (detection, segmentation, object tracking), OpenCV, image preprocessing
- **Programming & Data Handling:** Python (Pandas, NumPy), automation scripting
- **Data Pipelines & Automation:** End-to-end Python workflows (video → dataset → reports → GDBs)
- **Data Visualization & Analytics:** Matplotlib, Seaborn, predictive analytics
- **GIS & Geospatial Analysis:** ArcGIS Pro, GDB creation, spatial reporting, asset visualization
- **Model Evaluation:** Accuracy, Precision, Recall, F1 Score, MSE, RMSE, MAE
- **Annotation Tools:** CVAT, LabelImg, LabelMe – dataset preparation & quality control

CERTIFICATIONS & ACHIEVEMENTS

Certificate of Appreciation – Airloop

Awarded for sole execution and management of KSA Project 001, a flagship national-level ACI project.

Best Employee of the Quarter – Transconomy

Recognized for outstanding performance as a Data Extraction Operator.

EDUCATION

MCS – Master's in Computer Science

Islamia University of Bahawalpur | 2019–2021 | CGPA: 3.54

BSc

Islamia University of Bahawalpur | 2017–2019 | Marks: 523/800

REFERENCE

AVAILABLE UPON REQUEST